Hamidreza Ghasemi Damavandi

Principal Data Scientist | Ph.D., Electrical Engineering

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Personal Profile

Results-driven Principal Data Scientist with a Ph.D. in Electrical Engineering and over 8 years of experience architecting and deploying advanced machine learning solutions. Proven expertise in leading cross-functional teams to deliver scalable AI products, from generative AI and large language models (LLMs) to computer vision and deep learning. Adept at translating complex research into practical business applications with multiple cloud and AI certifications.

Technical Skills

Generative AI & LLMs	Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-tuning, LangChain, Hugging Face Transformers, OpenAI API, Vector Databases, Embeddings.
Machine Learning & Deep Learning	Supervised & Unsupervised Learning, Classification, Regression, Clustering, Time Series Analysis, Recommendation Systems, CNNs, RNNs/LSTMs, GANs, Transformers, PyTorch, TensorFlow, Keras, Scikit-learn.
Natural Language Processing	Sentiment Analysis, Topic Modeling, Named Entity Recognition (NER), Machine Translation, Text Summarization, Question Answering, Hugging Face, NLTK, SpaCy.
Computer Vision	Image Classification, Object Detection, Image Captioning, Face Generation, Landmark Recognition, OpenCV.
Cloud Computing & MLOps	AWS (SageMaker, Bedrock, Lambda, EC2, S3), Docker, Kubernetes, Jenkins, MLflow, FastAPI, Flask, Apache Spark.
Programming & Data Science	Python, SQL, R, Pandas, NumPy, Matplotlib, Seaborn, Git, GitHub, Data Structures, Algorithms.

Professional Experience

Arizona State University

Tempe, AZ

Principal Data Scientist

Dec 2020 – Present

- **Data Science & Machine Learning Initiatives:** Led the group's data science and machine learning efforts, including advanced model development, interpretability using SHAP, and integration into decision-making processes.
- **Student Prediction Models:** Developed and deployed a daily XGBoost model to predict students' likelihood of persistence, with SHAP-based explanations highlighting key drivers of low persistence, enabling proactive interventions at scale which impacted **100,000+ students**.
- **Gen AI Solution:** Designed and implemented an LLM-based solution leveraging AWS services (Bedrock, OpenSearch, Step Function, Lambda) with a Retrieval-Augmented Generation (RAG) architecture to support academic advisors in proactively identifying and assisting students in need of intervention.
- **MLOps:** Designed and maintained full end-to-end production pipelines – including model development, deployment, CI/CD, and monitoring – using Metaflow, AWS SageMaker, Step Functions, Lambda, GitHub, Terraform, and Jenkins. Reduced model deployment time from **hours to minutes** and ensured **99% pipeline reliability**.
- **Automation:** Automated the entire data science lifecycle – from feature engineering and model training to deployment and monitoring – by integrating Slack notifications and Amazon Q, enabling real-time alerts and reducing incident response time by **50%**.
- **Algorithm Development:** Developed and applied algorithms to optimize HigherEd processes, improving scheduling efficiency by **100%**.
- **Technical Mentorship & Presentations:** Onboarded 3+ new team members through training on production workflows and delivered knowledge-sharing presentations on advanced Machine Learning topics at institutional and external venues, reaching audiences of 50+.

Overstock

Sandy, UT

Senior Machine Learning Scientist

Mar 2020 – Dec 2020

- **Optimization Framework:** Developed a multi-variate constraint optimization framework using linear and integer programming to allocate discounts optimally for on-sale items.
- **Large-Scale Data Processing:** Handled datasets with millions of rows to mine correlations and support key business decisions. Built high-quality, object-oriented Python scripts, containerized with Docker, and orchestrated with Apache Airflow for production deployment.

Education

University of Iowa

Iowa City, IA

M.S. and Ph.D. in Electrical Engineering

Aug 2013 – Aug 2016

University of Tehran

Tehran, Iran

B.S. in Electrical Engineering

Aug 2009 – Jul 2013

Honors and Awards

- **Public University Entrance Exam** – **Ranked 236th** out of 310,000 applicants nationwide for B.Sc. admission, Summer 2009.
- **Azad University Entrance Exam** – **Ranked 5th** for B.Sc. admission, Summer 2009.
- **University of Tehran Admission** – Admitted to the **1st**-ranked university overall and **2nd** in Technology in Iran with a full four-year scholarship, Summer 2009.
- **NIW Green Card Award** – Received U.S. Permanent Residency via the **National Interest Waiver (NIW)** based on academic merit. A U.S. citizen since 2024.
- **Session Chair, AGU 2019** – Convened and chaired a session on Machine Learning Applications in Hydrology at the AGU Fall Meeting, 2019.
- **Graduate Research Scholarship** – Awarded full Graduate Research Assistantship at the University of Iowa, 2013.
- **Graduate Travel Award** – Received CGRER graduate travel award from the University of Iowa, Spring 2016.
- **High School Competition** – **Ranked 1st** in a comprehensive high school exam among schools in the eastern part of Tehran, Iran, Spring 2009.
- **High School Top Rank** – **Ranked 1st** at both high school and pre-university level among all students.

Research Interests

Artificial Intelligence • Machine Learning (Statistical & Scientific) • Deep Learning • Natural Language Processing • Large Language Models • Signal & Image Processing • Optimization • Applied Statistics

Relevant Courses

Machine Learning • Statistical Pattern Recognition • Optimization Techniques • Digital Signal Processing • Knowledge Discovery • Graph Algorithms & Combinatorial Optimization • Information Theory & Coding • Image & Video Compression • Stochastic Processes • Engineering Probability & Statistics • Signals & Systems • Engineering Mathematics

Selected Projects

- **Customized ChatBot Using OpenAI's LLMs:** Developed the end-to-end code to create a customized ChatBot leveraging OpenAI's LLM models. In particular, I utilized the Retrieval-Augmented Generation (RAG) technique to increase the accuracy of the LLM models. [code]
- **Face Generation:** Built a deep convolutional GAN (generative adversarial network) to generate fake images from real celebrity images as part of the Udacity Deep Learning course. [code]
- **Sentiment Analysis:** Implemented text translation and sentiment analysis using Hugging Face transformers. [code]
- **SLAM:** Implemented Simultaneous Localization and Mapping (SLAM) technique as part of the Udacity Computer Vision course to track the location of a robot in a 2D world in real-time and identify the locations of landmarks such as buildings, trees, rocks. [code]

- **Image Captioning:** Trained a CNN-RNN model to predict captions for a given image as part of the Udacity Computer Vision course. [code]
- **Facial Keypoint Detection:** Trained a Convolutional Neural Network using PyTorch to detect facial keypoints (mouth, nose, eyes) in an image as part of the Udacity Computer Vision course. [code]
- **DNN Speech Recognizer:** Built a deep neural network using PyTorch that functions as an element of an end-to-end automatic speech recognition pipeline as part of the Udacity NLP course. [code]
- **Machine Translation:** Implemented an encoder-decoder network using Recurrent Neural Networks for translating English sentences to French and vice versa as part of the Udacity NLP course. [code]
- **Part of Speech Tagging:** Built a hidden Markov model for part of speech tagging as part of the Udacity NLP course. [code]
- **Landmark Classification & Tagging for Social Media:** Built a Deep Convolutional Neural Network to classify different landmarks found on social media using PyTorch as part of the Udacity Deep Learning course. [code]

Selected Certifications

- **Johns Hopkins Certificate Program in Applied Generative AI** – JHU certification.
- **NVIDIA-Certified Associate: Generative AI/LLMs** – NVIDIA certification. [link]
- **LLMs & Text Generation (Udacity)** – Certificate earned through Udacity's AI Nanodegree program. [link]
- **Generative AI with LLMs (Coursera)** – Certificate earned via Coursera. [link]
- **Natural Language Processing (Udacity)** – Certificate earned via Udacity. [link]
- **Computer Vision (Udacity)** – Certificate earned via Udacity. [link]
- **Deep Learning (Udacity)** – Certificate earned via Udacity. [link]
- **AWS ML Specialty** – AWS certification. [link]
- **AWS Cloud Practitioner** – AWS certification. [link]
- **AWS AI Practitioner** – AWS certification. [link]
- **Software Engineering Essentials (Coursera)** – Certificate earned via Coursera. [link]
- **Data Structure and Algorithms (ASU)** – Certificate earned via ASU CareerCatalyst. [link]